COMPUTER AND COMMUNICATIONS ENGINEERING PROGRAM

Mission

The CCE program aims to help students explore their innate creativity and potential and endow them with the knowledge, skills, and abilities to pursue successful careers in computer and communication engineering and related fields; think critically in solving complex problems using modern tools and technologies; communicate and work effectively with diverse groups; learn more every day, and succeed in graduate studies in renowned institutions if they so choose.

Program Educational Objectives

The program's educational objectives are to enable graduates to:

1. Foster an environment that encourages excellence in endeavor.
2. Provide quality and professional education that prepares leaders.
3. Solve problems facing society and industry in Lebanon, the Middle East, and the world.
4. Engage in multidisciplinary research.

Student Outcomes

The Computer and Communication Engineering program adopts the learning outcomes of ABET, so our graduates will have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

6. An ability to develop and conduct appropriate experimentation, analyze, interpret data, and use engineering judgment to conclude.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Computer and Communications Engineering is accredited by the Engineering Accreditation Commission of ABET, https://www.abet.org, under the commission's General Criteria and Program Criteria for Computer and Communications and Similarly Named Engineering Programs.

Program Overviews

The Computer and Communications Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum tailored to market needs, facilities that meet the highest standards, top-notch faculty members who graduated from premier higher education institutions and are committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Computer and Communications Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 116 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or BE degree is conferred only if a student's earned Cumulative GPA, Major GPA, and Summative Learning Project grade are all at least 70.

The required credit hours are divided into three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in Computer and Communication Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	0	14	10	27	23
College Requirement	0	0	24	0	24	20
Program Requirement	46	0	21	0	67	57
Credits	49	0	59	10	118	100

BE in Computer and Communication Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	3	0	14	13	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	53	18	24	0	95	62
Credits	56	18	65	13	152	100

I. University Requirements (General Education)

The list of GE-required courses and their descriptions is presented on the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Computer and Communications Engineering (CCE) encompass 95 credits (67 credits for the BS degree) distributed as follows: 77 credits of mandatory courses (67 credits for the BS degree) and 18 credits (0 credits for the BS degree) of engineering technical elective courses for BE only.

A. Mandatory Requirements

The mandatory courses for the major and non-major CCE programs are listed in the table below.

Course #	Title	Credits	Pre-requisites
CCEE 221	Logic Design	3	
CCEE 221L	Logic Design Lab	1	CCEE 221
CCEE 216	Advanced Programming and Data Structure	3	CCEE 214
CCEE 310	Software Engineering	3	CCEE 214
CCEE 315	Database Management Systems	3	CCEE 214
CCEE 324	Computer Organization	3	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
CCEE 341	Communication Systems	3	CCEE 331
CCEE 354	Computer Networks	3	CCEE 214
CCEE 354L	Computer Networks Lab 1	1	Co-req.: CCEE 354
CCEE 400	BS Summative Learning Project	3	ENGL 217
CCEE 411	Web Programming	3	CCEE 214

CCEE 426	Design of Embedded Systems	3	Pre-req.: CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE 444	Antennas and Propagation	3	Pre-req.: ELEC 340
CCEE 444L	Antennas and Propagation Lab	1	Co-req.: CCEE 444
CCEE 447	Digital Communications	3	Pre-req.: CCEE 341
CCEE 447L	Communications Lab	1	Co-req.: CCEE 447
CCEE 449	Wireless Communications	3	Pre-req.: CCEE 447
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
CCEE 499	Co-op Experience	3	Pre-req.: ENGL 217
CCEE 528	Computer Architecture	3	Pre-req.: CCEE 324
CCEE 534	Signal Processing	3	Pre-req.: CCEE 331
CCEE 543	Mobile Communications	3	Pre-req.: CCEE 449
CCEE 564	Machine Learning and Data Mining	3	Pre-req.: CCEE 214; MATH 351 or BADM 250
CCEE 595A	BE Summative Learning Project 1	1	Pre-req.: ENGL 217
CCEE 595B	BE Summative Learning Project 2	3	Pre-req.: CCEE 595A
CCEE	CCEE Selected Elective	3	
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 320	Electronics	3	Pre-req.: ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ELEC 340	Engineering Electromagnetics	3	Pre-req.: PHYS 211

* Required for BS-bound students. BE students take technical electives instead.

B. Technical Electives

As part of the Bachelor of Engineering degree program in CCE, the student is required to take 18 credit hours of engineering technical elective courses. Students must be of senior standing to take engineering technical electives. Technical electives allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match engineering technical elective courses from different areas to gain broader exposure to the various CCE disciplines. Students should, in cooperation with their academic advisor, select a list of electives that best meet their needs and aspirations. The listed engineering technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Computer Software
2. Computer Hardware
3. Signal Processing
4. Communication Systems
5. Computer Networks
6. Artificial Intelligence

The list of additional available CCE electives is given in the table below:

Course #	Title	Credits	Pre-/Co-requisites
CCEE 498	Special Topics in Computer and Communication Engineering	3	
CCEE 597	Advanced Topics in Computer and Communication Engineering	3	Senior Standing
Computer Software			
CCEE 511	Mobile Application Development	3	Pre-req.: CCEE 214
CCEE 513	Operating Systems	3	Pre-req.: CCEE 214
CCEE 514	Advanced Web Programming	3	Pre-req.: CCEE 411
CCEE 515	Distributed Computing	3	CCEE 528
CCEE 516	Advanced Programming and Database Management Systems	3	Pre-req.: CCEE 315 or equivalent
Computer Hardware			
CCEE 520	Parallel Processing	3	Pre-req.: CCEE 221
CCEE 521	Hardware-Software Co-Design	3	Pre-req.: CCEE 221
Signal Processing			
CCEE 535	Digital Signal Processing	3	Pre-req.: CCEE 331
CCEE 536	Digital Image Processing	3	Pre-req.: CCEE 214
Communication Systems			
CCEE 540	Satellite Communication Systems	3	Pre-req.: CCEE 447
CCEE 542	Microwave Communications	3	Pre-req.: ELEC 340
CCEE 545	Advanced Mobile Communications	3	Co-requisite: CCEE 449
CCEE 546	Array Processing	3	Pre-req.: CCEE 544
CCEE 547	Optical Communications	3	Pre-req.: CCEE 447
Computer Networks			
CCEE 457	Cybersecurity Fundamentals	3	Pre-req.: CCEE 354
CCEE 552	Network Security	3	Pre-req.: CCEE 354
CCEE 553	Advanced Communication Networks	3	Pre-req.: CCEE 354
CCEE 554L	Computer Networks Lab 2	1	Co-req.: CCEE 354L
CCEE 555	Advanced Computer Networks	3	Pre-req.: CCEE 354
CCEE 555L	Computer Networks Lab 3	1	Pre-req.: CCEE 354L
CCEE 556	Network and System Administration	3	Pre-req.: CCEE 354
CCEE 556L	Computer Networks Lab 4	1	Pre-req.: CCEE 555L
Artificial Intelligence			
CCEE 561	Computer Vision	3	Pre-req.: CCEE 214
CCEE 563	Robotics	3	
CCEE 566	Natural Language Processing	3	Pre-req.: CCEE 214
CCEE 567	Optimization	3	Pre-req.: MATH 215 or equivalent
CCEE 568	Big Data and Analytics/Big Data System	3	Pre-req.: CCEE 564 or equivalent

RHU is a CISCO Academy. As such, students will be awarded CISCO certificates after completing the required Computer Networks Lab 1. In addition, students can take any three of the following labs, which will count as one technical elective.

Course #	Title	Credits	Pre-/Co-requisites
Computer Networks			
CCEE 554L	Computer Networks Lab 2	1	Co-req.: CCEE 354L
CCEE 555L	Computer Networks Lab 3	1	Pre-req.: CCEE 354L
CCEE 556L	Computer Networks Lab 4	1	Pre-req.: CCEE 555L

Undergraduate students can also take 600-level courses, provided that their cumulative GPA is above 80 and the instructor approves.

Students may up to 9 credits of (400-level or above) engineering technical electives from other programs,.

C. Summative Learning Project (SLP)

Students must complete a 3-credit-hour course for BS (4 credits for BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to innovate a promising idea.

D. Co-op Experience

Students must complete 8 weeks of practical training working in an area related to their field of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on the student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

Code numbers designate the courses offered in the CCE program in the form of (CCEE abc), where:

a	Designates the year (level)
b	Designates the focus area as follows: 0: General; 1: Computer Software; 2: Computer Hardware; 3: Signal Processing; 4: Communication Systems; 5: Computer Networks; 6: Artificial Intelligence
c	Designates the course sequence in a focus area

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in CCE. The study plan serves as a roadmap to facilitate students' smooth progression toward graduation.

Course #	Title	Credits	Pre-/Co-requisites
Year 1, Fall Semester (14 Credits)			
CCEE 214	Introduction to Programming	3	
ARAB 212	Arabic Language & Communication	2	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 211	Physics: Electricity and Magnetism and Lab	3	
Year 1, Spring Semester (15 Credits)			
CCEE 216	Advanced Programming and Data Structure	3	CCEE 214
CCEE 221	Logic Design	3	
ELEC 200	Introduction to Electrical and Computer Engineering	3	
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
Year 1, Summer Semester (10 Credits)			
CCEE 221L	Logic Design Lab	1	CCEE 221
MATH 311	Linear Algebra and Applications	3	
	Science Elective	3	
	Humanities/ Fine Arts Elective I	3	
Year 2, Fall Semester (17 Credits)			
CCEE 324	Computer Organization	3	CCEE 221
CCEE 331	Signals and Systems	3	MATH 216
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
ENGL 217	Professional English Communication	3	ENGL 210
CCEE 354	Computer Networks	3	Co-req.: CCEE 221
CCEE 354L	Computer Networks Lab 1	1	Co-req.: CCEE 354
Year 2, Spring Semester (16 Credits)			
CCEE 310	Software Engineering	3	CCEE 214
CCEE 315	Database Management Systems	3	CCEE 214
CCEE 341	Communication Systems	3	CCEE 331
ELEC 340	Engineering Electromagnetics	3	PHYS 211
MATH 314	Ordinary Differential Equations	3	MATH 216
	Community and Sustainability Elective	1	
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economy and Management	3	

MATH 421	Numerical Analysis	3	MATH 311
	Social Science Elective I	3	
Year 3, Fall Semester (17 Credits)			
CCEE 411	Web Programming	3	Co-req.: CCEE 214
CCEE 447	Digital Communications	3	CCEE 341
CCEE 447L	Communications Lab	1	Co-req.: CCEE 447
CCEE 426	Design of Embedded Systems	3	CCEE 221
CCEE 426L	Design of Embedded Systems Lab	1	Co-req.: CCEE 426
CCEE	Selected CCEE Elective	3	
MATH 351	Probability and Statistics	3	MATH 216
Year 3, Spring Semester (17 Credits)			
CCEE 444	Antennas and Propagation	3	ELEC 340
CCEE 444L	Antennas and Propagation Lab	1	Co-req.: CCEE 444
CCEE 449	Wireless Communications	3	Co-req.: CCEE 447
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
MATH 210	Discrete Mathematics	3	MATH 215
	Engineering Technical Elective 1 or BS Project*	3	
Year 3, Summer Semester (3 Credits)			
CCEE 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Computer and Communication Engineering.			
Year 4, Fall Semester (16 Credits)			
CCEE 534	Signal Processing	3	CCEE 331
CCEE 528	Computer Architecture	3	CCEE 324
CCEE 595A	BE Summative Learning Project 1	1	ENGL 217; Senior Standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective 2	3	
	Engineering Technical Elective 3	3	
Year 4, Spring Semester (18 Credits)			
CCEE 543	Mobile Communications	3	Co-req.: CCEE 449
CCEE 564	Machine Learning and Data Mining	3	CCEE 214; MATH 351 or BADM 250
CCEE 595B	BE Summative Learning Project 2	3	CCEE 595A
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take CCEE 400 – Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
9 credits restricted Departmental Electives
9 credits in any Engineering discipline at the 400 level or above.